🔑 **Answer Key — Teacher Copy · Lesson 7-3**

MSTAR PRACTICE WORKSHEET

Lesson 7-3: Understand Absolute Value of Rational Numbers

Grade 6 Mathematics · Standard 6.NOS.8

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: D

What is the absolute value of -9?

A -9

B 0

C 1

D 9

Absolute value is the distance from zero. -9 is 9 units from zero, so |-9| = 9.

Question 2 SELECTED RESPONSE — CORRECT: D

Which integer has the GREATEST absolute value: -3, 7, -5, 2?

A -5

B -3

C 2

D 7

|-3| = 3, |7| = 7, |-5| = 5, |2| = 2. The greatest absolute value is 7.

Question 3 SELECTED RESPONSE — CORRECT: C

What is |0|?

A 1

B -1

C 0

D There is no answer

Zero is 0 units from itself on the number line, so |0| = 0.

Question 4 SELECTED RESPONSE — CORRECT: B

What is |−2.5|?

A 0

B 2.5

C −2.5

D 25

Absolute value is distance from zero, and −2.5 sits 2.5 units from zero, so |−2.5| = 2.5.

Question 5 SELECTED RESPONSE — CORRECT: B

What is |−3/4|?

A 0

B 3/4

C −3/4

D 4/3

|−3/4| = 3/4. Absolute value is distance from zero, and −3/4 sits 3/4 of a unit from 0 — fractions work exactly like integers here.

Question 6 SELECTED RESPONSE — CORRECT: A

Which statement about absolute value is TRUE?

A |-4| = |4| because both are 4 units from zero

B |-4| = -4 because the number is negative

C |-4| = 0 because negatives cancel out

D |4| > |-4| because positive is always bigger

Absolute value measures distance from zero. Both -4 and 4 are 4 units from zero, so |-4| = |4| = 4.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

What is the value of |–12|?

A 0

B 1/12

C 12

D –12

Absolute value is distance from zero, and distance is never negative. –12 sits 12 units from zero, so |–12| = 12.

- **A:** The bars ask how FAR from zero, not the difference with itself. Negative 12 sits 12 units away.
- **B:** The bars mean distance from zero, not a reciprocal.
- **D:** Absolute value is a distance from zero, and a distance is never negative.

Part B — correct answer: D

Why can an absolute value never be negative?

A Negative signs cancel out automatically.

B The bars change every number to a positive.

C Absolute value always adds the number to itself.

D It measures distance from zero, and a distance is never negative.

|a| is defined as how FAR a is from zero, regardless of direction. Distance has no direction, so it is always zero or positive.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Bigger Absolute Value, Bigger Number?

> Owen compared −8 and −3. He wrote: 'Since |−8| = 8 and |−3| = 3, and 8 > 3, then −8 > −3.'

Explain Owen's misconception and state the correct comparison.

> **Model answer:** −8 < −3. Absolute value tells you how far a number is from zero, not which is greater. A negative number with a larger absolute value sits further left, so it is SMALLER. Owing $8 is worse than owing $3.

2 points	Student explains that absolute value measures distance and ignores direction, that on the number line −8 lies further LEFT than −3, and concludes −8 < −3.
1 point	Student states −8 < −3 without explaining why absolute value cannot be used to compare.
0 points	Student agrees with Owen or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A submarine is at -150 meters and a bird is flying at +150 meters. A student says the submarine is closer to sea level because -150 is less than 150. Is the student correct? Explain using absolute value.

> **Model answer:** The student is not correct. Absolute value measures distance from sea level, and |-150| = 150 while |150| = 150, so the submarine and the bird are exactly the same distance from 0. The comparison -150 < 150 describes position — below versus above sea level — not closeness to it.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.